

Sunny Zhang

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EDUCATION

University of Toronto, St. George

Honours BSc, Double Major in Computer Science and Mathematics

2025 – 2029

GPA: 3.7/4.0

EXPERIENCE

AI Research Intern

Jun 2026 – Present

Algoverse AI Research Program

- Lead author of a preprint showing LLM manager agents conceal competitive pressure in their language while acting on it, allocating up to an entire capital pot to one agent.
- Built a 4-agent LLM trading firm in Python (Llama-3.3-70B, 120 episodes, 2,400 agent decisions) that labels misconduct from simulator ground truth.
- Proved transcript-based monitoring cannot detect the failure ($p < 10^{-6}$), evidence that AI oversight must instrument what agents do, not only what they say.

PROJECTS

LaunchPilot – Hack Canada 2x Track Winner | *Next.js, TypeScript, FastAPI, Postgres*

- Built a supervised 3-stage agent pipeline with persistent threads and memory snapshots, gated by Auth0-scoped permissions and human approval before any outreach sends.
- Shipped a 27-endpoint FastAPI REST API on Postgres with SQLAlchemy models and Alembic migrations, plus a 13-route Next.js App Router frontend in TypeScript.
- Containerized Postgres, migrations, and the API with Docker Compose.

Feedback or Annealing? | *Python, NumPy, Numba, Matplotlib*

- Derived a geometric envelope for residual-driven aggregation from Bellman span contraction, creating an open-loop control that isolates feedback from the annealing schedule it induces.
- Evaluated matched residual, geometric, and fixed-width methods on a 9,261-state MDP across 3 update mixes, 20 paired seeds, and 5 compute budgets; feedback improved error by at most 0.3% and worsened it by up to 14.5%.
- Showed that annealing changed error by up to 23.5% relative to fixed resolution while all final feedback comparisons remained within a preregistered ± 0.02 equivalence region.

Parallel State Aggregation | *Python, NumPy, Numba, Matplotlib*

- Built a multithreaded benchmark of value iteration utilizing numba and adaptive state aggregation across 40,000–1,000,000-state MDPs, 1–10 threads, and paired seeds.
- Found value iteration reached the fixed ℓ_∞ error target 1.3–2.2 $\times$ faster across all 5 configurations; traced scaling limits to memory bandwidth and serial rebinning and exposed 99.67% uncounted work.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL

Machine Learning and Data: PyTorch, Gymnasium, NumPy, Pandas

Framework & Libraries: FastAPI, Next.js, Numba, Flask, D3

Tools and Platforms: Git, Docker, Supabase, Hugging Face, Vercel